amazon rme

AMAZON BASE BUILDING MAINTENANCE STANDARD

GLOBAL CENTRAL RME STANDARDS & COMPLIANCE

Task Title	HVAC SYSTEMS, AIR ROOFTOP UNITS-OUTDOOR - 9
APM Object	HVAC.RTU
APM EC	AMZ.HVAC.RTU.GRP
BBM Handbook Title	Air Rooftop Units (RTUs)
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure These tasks are based on industry guidance, SME knowledge and HIE learning. Please ensure vendor specific requirements are followed. To suggest improvements please raise a SIM SIM - Link to SIM template

Last Update: (Version)	Jul-23	**Geo Applied:**	Intl APM Instance	**F-Gas**	Yes	**Activies may be covered by No-Go restrictions:**	YES	**Special Certificaion Required:**	Hygiene certification required in some countries (e.g. VDI 6022)	**Delivered By:**	Carried out by trained on site technician certified in the electrical field	**OCB:**	BBM
Evidence for CT:	YES	**Pricing template Position:**	9	**Related CT Areas:**	CT.AIR.INDOOR CT.HVAC.12KW CT.HVAC.FGAS	**Assets included in this PM:**	ROOFTOP UNIT OUTDOOR	**Schedule With:**	HVAC.HMD HVAC.BLR (if installed)				

Description of System / Technology (According to BBM Handbook)

Roof Top Units (or RTUs) are a packaged unit that is a typical HVAC system type seen across Amazon FC's. RTUs contain all the components needed to provide conditioned air in one concise unit with a single unit being able to provide comfort cooling, heating, supply of an adjustable volume of fresh air, as well as free cooling.

Additional Information

All sites with the above listed Base Building Asset should conduct routine maintenance and inspections according to this standard PM package. The Amazon maintenance standard supports compliance, reduces safety risks,improves asset reliability & efficiency.
Some maintenances captures several assets within one PM package -> Ensure that corrective/reactive maintenance for assets where proactive maintenance is contained under another asset is created against the original asset.
Note: the yearly PPM should be planned before the summer season starts.
Refrigerant part -> Ensure that leakage and any other requirements according to F Gas Regulation are followed

Task Index	Component	Tasks	Tasks Nested for frequencies > weekly									
			Daily	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	36 Monthly	48 Monthly	60 Monthly
1	General	Is the technical documentation & blueprints of installed system available & complete						PM				
2	Controller	Is the unit in operation & no alarms indicated - check unit control cabinet					PM					
3	Controller	Are any settings in manual mode -clarify the reason & set back to automatic mode										
4	Control cabinet	Visual check of the control cabinets inside & outside for damage, debris & missing components						PM		PM		
5	Control cabinet	Check circuit diagrams for completion						PM		PM		
6	Control cabinet	If anything is missing or unclear in the circuit diagrams create a WO						PM		PM		
7	Control cabinet	Are all cables properly routed						PM		PM		
8	Control cabinet	Are all cables, terminal blocks & devices properly labeled or marked						PM		PM		
9	Control cabinet	Test all safety relevant switches						PM		PM		
10	Control cabinet	Were any obvious changes made (e.g. rewiring, hanging cables)						PM		PM		
11	Control cabinet	Visual check for the protection against direct contact with live part						PM		PM		
12	Control cabinet	Tight cable connections in the control cabinets & main supply connection boxes						PM		PM		
13	Control cabinet	Are all metal parts connected to the protection earth (ground)						PM		PM		
14	Control cabinet	Replace the air filters of the cabinet & check operation of cooling fan						PM		PM		
15	Control cabinet	Are all contacts of the compressor relais colsing & opening properly						PM		PM		
16	RTU outside check	Is the unit freely accessible & maintainable			SC	SC	PM	PM		PM		
17	RTU outside check	Is the unit in operation & no alarms indicated - check unit control cabinet			SC	SC	PM	PM		PM		
18	RTU outside check	Check unit air intakes & outlets for corrosion & damage, contamination & condensation			SC	SC	PM	PM		PM		
19	RTU outside check	Is there any unusual noise or vibration that could be caused by a faulty unit component			SC	SC	PM	PM		PM		
20	RTU outside check	Are there any visual damage, missing- or loose components at the unit			SC	SC	PM	PM		PM		
21	RTU outside check	Is water dropping out of the unit			SC	SC	PM	PM		PM		
22	RTU outside check	Check of the drain & drain trap (e.g. ball siphon) connected to the unit, chek for correct function & damage			SC	SC	PM	PM		PM		
23	Air Handling Part filter	Check filters for unacceptable contamination, damage, leaks (can the air bypass the filter medium?)			SC	SC	PM	PM		PM		
24	Air Handling Part filter	Measure differential pressure of all filter stages by using adequate tools			SC	SC	PM	PM		PM		
25	Air Handling Part filter	List the differential pressures in the notes			SC	SC	PM	PM		PM		
26	Air Handling Part filter	Schedule filter change when 80% of the capacity is reached			SC	SC	PM	PM		PM		
27	Air Handling Part filter	(An increased filter change is ok due to environmental impact)						PM		PM		
28	Air Handling Part filter	In the notes, please enter the number of filter changes per filter stage within the year						PM		PM		
29	Air Handling Part Fan	Change belts and check pulleys at belt driven fans						PM		PM		
30	Air Handling Part Hygiene & damages	Check droplet eliminator and tray for contamination e.g. mold, corrosion & damage				SC	PM	PM		PM		
31	Air Handling Part Hygiene & damages	Check cooling coil & condensate tray for contamination, corrosion & damage				SC	PM	PM		PM		
32	Air Handling Part Hygiene & damages	Clean unit air intakes and outlets						PM		PM		
33	Air Handling Part Hygiene & damages	Clean droplet eliminator & tray and check for contamination e.g. mold, corrosion & damage						PM		PM		
34	Air Handling Part Hygiene & damages	Clean cooling- & heating coil, as well as condensate tray using professional cleaning materials						PM		PM		
35	Air Handling Part Hygiene & damages	Check for contamination, corrosion & damage						PM		PM		

36	Air Handling Part Hygiene & damages	Clean fans & impeller and check for contamination, corrosion & damage						PM		PM			
37	Air Handling Part Hygiene & damages	Clean Chamber surfaces inside the unit and check for contamination, corrosion & damage						PM		PM			
38	Air Handling Part Hygiene & damages	Visual check of heat recovery for contamination, corrosion & damage					PM	PM		PM			
39	Air Handling Part Hygiene & damages	Visual check of heating coil for contamination, corrosion & damage					PM	PM		PM			
40	Air Handling Part Hygiene & damages	Visual check of silencers for contamination, corrosion & damage						PM		PM			
41	Air Handling Part Heat exchange	Functional check of frost protection thermostat						PM		PM			
42	Air Handling Part Heat exchange	Check the electrical heater for contamination, corrosion & damage						PM		PM			
43	Air Handling Part Heat exchange	Check the electrical heater according to local electrical regulations						PM		PM			
44	Air Handling Part Heat exchange	Check directly heated heat exchanger for tightness						PM		PM			
45	Air Handling Part Heat exchange	Vacuum all remaining heat exchanger carefully						PM		PM			
46	Air Handling Part Heat exchange	Check of reheater & cooler installed in the duct system for contamination, corrosion & damage						PM		PM			
47	Air Handling Part Heat exchange	Change filter of reheater & cooler installed in the duct system						PM		PM			
48	Air Handling Part Heat exchange	Measure differential pressure of all heat exchanger, compare with design and schedule professional cleaning if differential pressure is 30% higher than designed -> use professional tool								PM			
49	Air Handling Part Heat Recoery wheel	check the heat recovery wheel for correct functionality & for damage						PM		PM			
50	Air Handling Part duct system	Check air outlets & intakes of the duct system for debris, contamination, corrosion & damage						PM		PM			
51	Air Handling Part duct system	Choose 5 locations and add the locations into the notes Check inner duct system checks for contamination, corrosion, condensation & dirt						PM		PM			
52	Air Handling Part duct system	Check Textile air hose for damage, contamination & dirt						PM		PM			
53	Air Handling Part duct system	Schedule exchange or washing of the textile air hose if necessary						PM		PM			
54	Air Handling Part duct system	Underground components - check for contamination, corrosion & dirt deposit inside of the duct						PM		PM			
55	Air Handling Part Measurements	Measure the power consumption of the Fans in full speed with open dampers						PM		PM			
56	Air Handling Part Measurements	Add the values into the notes indicating the fans & compare to design specifications of the fan						PM		PM			
57	Air Handling Part Measurements	Measure the main discharge & return air volume flow - add airflow and position into the notes						PM		PM			
58	Air Handling Part Measurements	Compare measured airflow with design specification						PM		PM			
59	Air Handling Part Measurements	Measure the main discharge & return air temperature						PM		PM			
60	Air Handling Part Measurements	Compare measured temperature with value in the BMS, add temperature and create a FWO under BMS in case of deviation						PM		PM			
61	Refrigerant Part outside check	Is there any unusual noise or vibration that could be caused by compressors or condenser fans					PM	PM		PM			
62	Refrigerant Part outside check	Is the condenser coil free from dirt, damage & corrosion Schedule cleaning if airflow through coil is impaired			SC	SC	PM	PM		PM			
63	Refrigerant Part outside check	Are there any visual damage, missing- or loose components or corrosion at the unit					PM	PM		PM			
64	Refrigerant Part outside check	Is there any deposit in the drip tray (system fluid)					PM	PM		PM			
65	Refrigerant Part outside check	Is there oil deposit visible at the connections of the pipe system					PM	PM		PM			
66	Refrigerant Part outside check	Is the pipe insulation free from damage					PM	PM		PM			
67	Refrigerant Part Refrigerant leakage	Is an appropriate record for the refill of refrigerant physically available						PM		PM			
68	Refrigerant Part Refrigerant leakage	Add to notes the refill of refrigerant during the last year in KG (year in maintenance interval, not calendar year)						PM		PM			
69	Refrigerant Part Refrigerant leakage	Add to notes details of reasaon for refrigerant refill						PM		PM			
70	Refrigerant Part functional cleaning	Clean condenser coil with state-of-the-art technology (professional)						PM		PM			
71	Refrigerant Part functional cleaning	Clean condenser fan blades (& housing in case of dirt deposit)						PM		PM			
72	Refrigerant Part functional cleaning	Carry out a functional cleaning of all reachable components						PM		PM			
73	Refrigerant Part Measurements	Connect the refrigerant manometer & temperature sensors (always compare with controller data)						PM		PM			
74	Refrigerant Part Measurements	Are the refrigeration pressures and refrigerant temperatures correct						PM		PM			
75	Refrigerant Part Measurements	Is the expansion valve working properly & superheat OK						PM		PM			
76	Refrigerant Part Measurements	Is the undercooling temperature ok						PM		PM			
77	Refrigerant Part Measurements	Are the condenser & evaporator temperatures ok						PM		PM			
78	Refrigerant Part Measurements	Measure the power consumption of the compressors while operating on 100%Compare the power consumption with the manufacturer specification - react in case of deviation						PM		PM			
79	Refrigerant Part Measurements	Measure the power consumption of the condenser fans while operating on 100% Compare the power consumption with the manufacturer specification - react in case of deviation						PM		PM			
80	Refrigerant Part Measurements	Add the operational hours of all compressors into the notes						PM		PM			
81	Refrigerant Part Measurements	Identify & eliminate the root cause if there are big differences in operational hours						PM		PM			
82	Refrigerant Part Measurements	Are the compressors switched in the correct order						PM		PM			
83	Refrigerant Part Measurements	Measure the oil pressure & adjust if required						PM		PM			
84	Refrigerant Part Measurements	Is the oil temperature/presuure before & after refrigerant filter ok						PM		PM			
85	Refrigerant Part Measurements	Is the compressor oil level ok						PM		PM			
86	Refrigerant Part Measurements	Conduct oil analyses & schedule actions if required						PM		PM			
87	Refrigerant Part Compressor & Safety	Does the low-pressure switch work properly						PM		PM			
88	Refrigerant Part Compressor & Safety	Does the high-pressure switch work properly						PM		PM			
89	Refrigerant Part Compressor & Safety	Simulate safety switches of the system (e.g. thermostats, timers, etc.)						PM		PM			
90	Refrigerant Part Compressor & Safety	Is the crankcase heater working properly (attached to compressor)						PM		PM			
91	Refrigerant Part Compressor & Safety	Is the oil separator working properly						PM		PM			
92	Refrigerant Part Compressor & Safety	Check if there is humidity in the refrigeration system						PM		PM			
93	Refrigerant Part Compressor & Safety	Check refrigerant flow through show glass						PM		PM			
94	Refrigerant Part Compressor & Safety	Does the condensate tub heater work properly						PM		PM			
95	Refrigerant Part Compressor & Safety	Check correct funcitonlity of coil defrost						PM		PM			
96	Refrigerant Part Compressor & Safety	Screw Compressor - return valve check						PM		PM			

97	Refrigerant Part Compressor & Safety	Lubricate all moving parts and bearings if reachable						PM		PM		
98	Final Check	Check the O&M manual and carry out additional required tasks						PM		PM		

Change Log

July 2023 - First issue in new format. Simplification of task scope, alignment with CT.
June 2024 - Moved "Humidifier tasks" to HVAC.HMD PM.